



Meaning of «Micro-Doppler»

(<https://www.mathworks.com/help/phased/ug/introduction-to-micro-doppler-effects.html>)

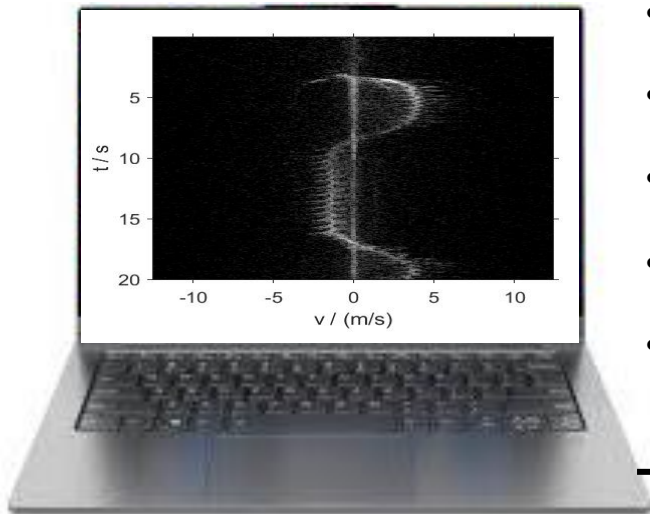
A moving target introduces a frequency shift in the radar return due to Doppler effect. However, because most targets are not rigid bodies, there are often other vibrations and rotations in different parts of the target in addition to the platform movement. For example, when a helicopter flies, its blades rotate, or when a person walks, their arms swing naturally. These **micro scale movements** produce additional Doppler shifts, referred to as **micro-Doppler effects**, which are useful in identifying **target features**.

Some questions in CW-Radar signal processing

- Can the steps of a person be detected in the radar signal?
- Is a human / non-human presence detector realizable?
- How fast rotate the blades of a fan / drone / ...?

Setup for Micro-Doppler Evaluation

Matlab (or Python)



- sampling (audio) signals with $f_s = 48$ kSps (or 44.1 kSps)
- resampling to 4, 6, 8, 9.6, 12 or 16 kSps according need
- record to wav-file & offline-computation of «spectrogram»
- alternatively, online-computation (live-mode)
- option: synch with video sequence

ASIO driver
USB

Notebook (or Smartphone)

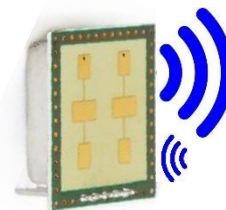


Instruments-Inputs (not XLR!)

$Q(t)$

$I(t)$

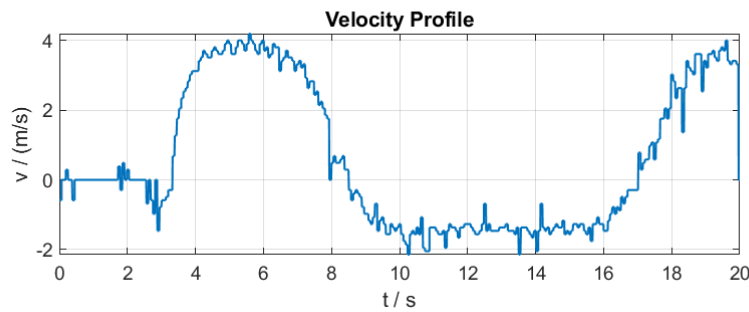
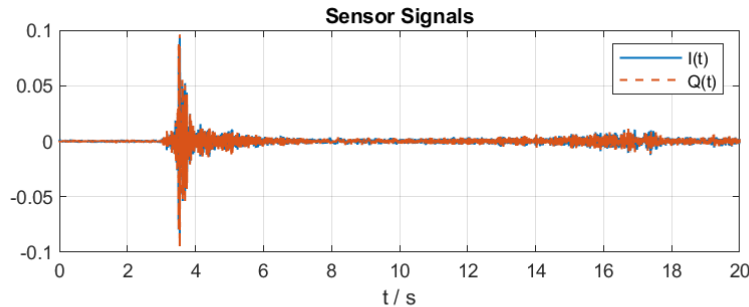
5V



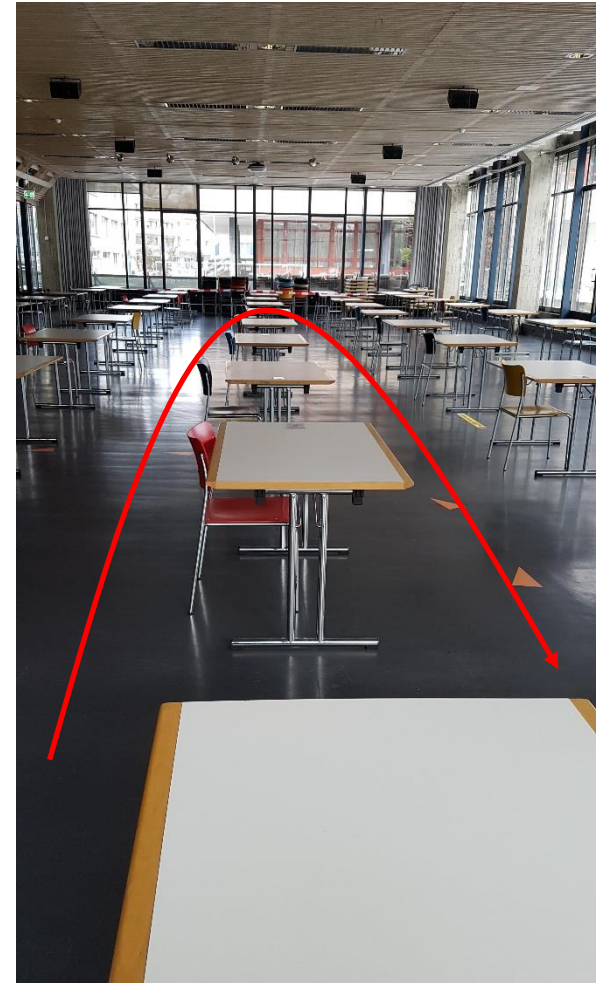
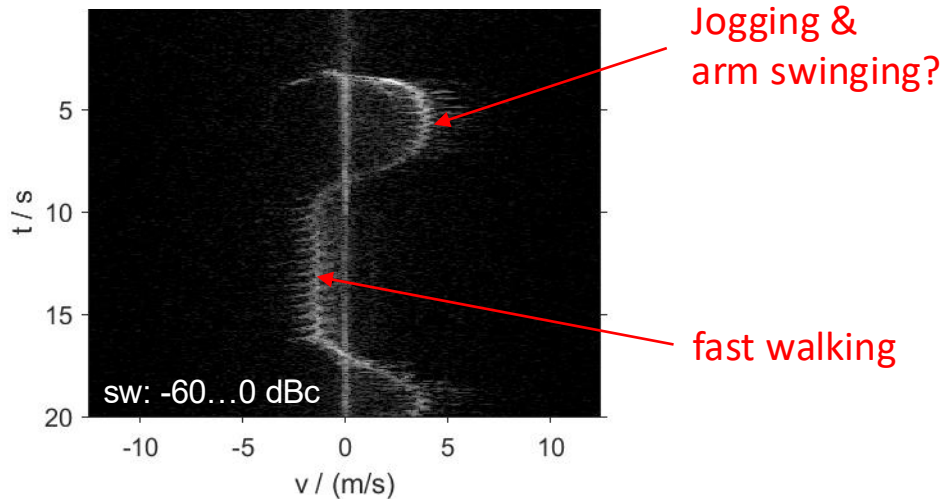
Sensor-Board
(without MCU-Board)



Results of a field test



$\Delta v = 0.1 \text{ ms}$
 $|v| > 0.2 \text{ ms}$



Radar K-LC6, 60 dB Gain, USB-Audio-Interface
($f_s = 4 \text{ kSps}$, $N_{\text{FFT}} = 256$, hanning-window)